Hall Ticket Number:

MR-22

Code: B325CM2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2025/January 2026

Software Engineering

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List the types of other software process models.	1	1	1	2
1. b)	List the types of software myths.	1	1	1	2
1. c)	Explain functional requirements.	1	1	2	2
1. d)	Explain user requirements in detail.	1	1	2	2
1. e)	List kinds of behavioural and object models.	1	1	3	3
1. f)	Explain requirement review.	1	1	3	2
1. g)	Describe white box testing.	1	1	4	1
1. h)	Distinguish between testing and debugging.	1	4	4	2
1. i)	What is FTR?	1	1	5	4
1. j)	Define risk components.	1	1	5	4

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain "Evolutionary Process Flow" with a neat sketch.	5	2	1	3
2. b)	Explain in detail about Scrum.	5	4	1	5
	OR				
2. c)	Explain framework activities with a neat diagram.	5	2	1	2
2. d)	Explain briefly the Generic Process Model.	5	2	1	3
3. a)	How requirements are classified? List them with an example for each.	5	6	2	2
3. b)	Classify the following as functional /non-functional requirements for a banking system. i. Verifying bank balance ii. Withdrawing money from a bank iii. Completion of transactions in less than one second. iv. Extending the system by providing more tellers for the customers	5	3	2	5
	OR				
3. c)	Analyze the requirement engineering process and how the requirements are managed.	5	4	2	2
3. d)	What is requirement elicitation? Briefly describe the various activities performed in requirements elicitation with an example of a watch system that facilitates setting time and alarm and assessment.	5	2	2	2
4. a)	Analyze briefly the structural system analysis in detail.	5	4	3	3
4. b)	What is the purpose of data flow diagrams? What are the notations used for the same?	5	4	3	3
	OR				
4. c)	Develop an online railway reservation system, which allows the user to select book/cancel tickets using net banking/credit/debit cards.	5	6	3	5
4. d)	Draw and explain the use case diagram for an ATM system in requirement elicitation.	5	3	3	5

P.T.O.

5. a)	Generalize your view about different varieties of testing.	5	4	4	3
5. b)	Discuss black box testing in a detailed view.	5	3	4	3
	OR				
5. c)	Explain about metrics for maintenance.	5	2	4	1
5. d)	Discuss the process of debugging.	5	2	4	1
6. a)	Discuss about reactive Vs Pro-active risk strategies.	5	3	5	1
6. b)	Create a risk table for the project that if you the project manager for a major software company. You have been asked to lead a team that's developing "next generation" "word processing software"	5	6	5	4
	OR				
6. c)	Define the Risk Projection Process.	5	3	5	4
6. d)	Discuss about Software Reviews.	5	4	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome